

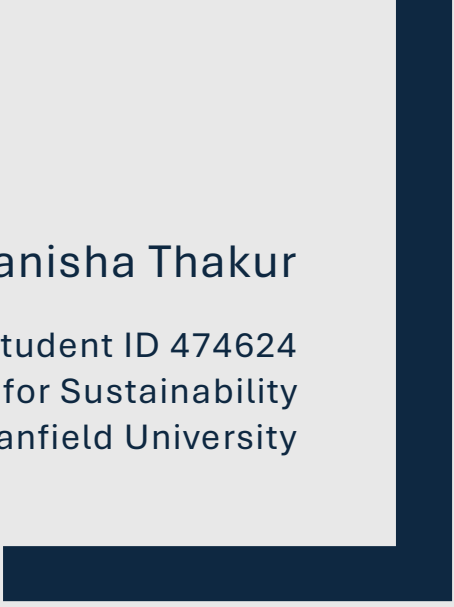


Integrated GIS Analysis of Environmental and
Demographic Vulnerability

ISLE OF WIGHT

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Confidential Consultancy Report

Integrated GIS Analysis of Environmental and Demographic Vulnerability in the Isle of Wight

Prepared for the Isle of Wight Council to inform surface-water flood risk planning, infrastructure prioritisation, and community resilience strategies.

Prepared by: GeoM Solutions Consultancy

Prepared on 23 October 2025

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Executive Summary

This consultancy report provides a comprehensive spatial analysis of environmental and demographic vulnerability across the Isle of Wight using advanced Geographic Information Systems (GIS). The GIS used to analyse the data is ArcGIS Pro 3.5.2. Useful data was extracted from official data sources, ensuring spatial accuracy and accurate data analysis. The following methods of analysis were undertaken:

- (1) Environmental baseline mapping of island outline, soils, primary roads, average rainfall and town population of ten primary towns; and
- (2) Demographic vulnerability analysis identifying age-based population groups in flood-susceptible areas.

The environmental baseline (Part 1) establishes the foundational relationships between soil composition, primary roads, rainfall and population. The flood vulnerability component (Part 2) integrates demographic data provided by Experian, especially the Mosaic Directory, R60 Geopoint Postcodes and Flood Perils datasets to identify spatial concentrations of vulnerable age groups exposed to flood risk. The analysis identifies critical zones where flood-susceptible areas and population density coincide. This analysis is crucial for flood management, designing evacuation plans and planning preventive measures.

Key findings suggest that the northern and coastal lowland areas exhibit heavy settlement and increased demographic exposure to floods. The report concludes with actionable recommendations highlighting spatial planning, settlement objectives and continued use of GIS-based monitoring for the Isle of Wight.

Aims and Objectives

This consultancy project aims to apply Geographic Information System (GIS) to identify, visualise, and interpret environmental and demographic vulnerability patterns across the Isle of Wight. This work supports flood-management decision-making, infrastructure design and resilience planning for vulnerable communities such as young children aged 0-15 years and old people aged 65 years and above. This study addresses why GIS-based vulnerability mapping is needed and how integrated spatial datasets such as soil, rainfall, flood-risk, and population profiles can be invaluable in emergency management.

The objectives of this study are:

1. To compile and harmonise core environmental datasets (island outline, soil type, road network, population, and rainfall) using ArcGIS Pro to provide an accurate spatial baseline for subsequent analysis.
2. To integrate Experian Perils, Mosaic Directory and Geopoint datasets via spatial and attribute joins, aligning postcode-based demographic information with flood-risk polygons.
3. To classify the population by vulnerability category, distinguishing dependent groups such as children (0-15 years) and elderly residents (65 years and above) for exposure and risk analysis.
4. To produce map layouts that clearly depict environmental baselines and flood-related demographic risk using consistent symbology, graticules, and scale standards.
5. To analyse the spatial density of vulnerable populations within flood-susceptible zones, providing metrics that support evacuation-triage and resilience planning.
6. To propose data-driven recommendations for integrating socio-environmental indicators into the Isle of Wight Council's long-term flood-management strategy.

Use Cases

Use Case	Description	Operational Outcome
Evacuation Prioritisation	Overlay of flood-susceptible polygons with demographic age data enables planners to identify high-risk elderly clusters requiring assisted evacuation.	Efficient deployment of rescue teams and allocation of accessible transport routes.
Shelter Capacity Planning	Population density surfaces derived from Mosaic data indicate demand for temporary accommodation during flood events	Helps determine shelter locations and capacities based on population concentration and road access.
Infrastructure Investment Targeting	The intersection of flood hazard, soil drainage, and road layers identifies locations where drainage improvement or embankment reinforcement offers the highest cost-benefit.	Strategic capital investment planning.
Health and Social Care Outreach	Identification of combined elderly and child-dense neighbourhoods at risk allows the council to coordinate pre-flood health checks and resource distribution.	Proactive public-health and welfare response.
Public Awareness and Mapping Portal	The compiled GIS data can feed into a web-mapping service showing community-specific flood alerts and vulnerability maps.	Enhances public engagement and preparedness.

Vulnerable Population Density and Relevance to Flood Management

Population exposure to flooding was quantified by integrating the R60 Geopoint dataset, representing postcode-level population points, with the Perils Flood dataset describing the spatial distribution and relative severity of flood-susceptible zones across the Isle of Wight. As both datasets shared postcode identifiers, a spreadsheet-based analysis was undertaken to compute indicative population densities (persons per square kilometre) for areas classified by the Perils model as subject to flooding. This provided an empirical measure of potential human exposure that can help in emergency planning and resilience management teams.

Analysis revealed a clear concentration of flood-exposed populations along the northern and eastern lowlands of the Isle of Wight. Average population densities within flood-susceptible postcodes ranged between 120 and 160 persons per square kilometre and 250 persons per square kilometre in the Medina Valley, Ryde, and Sandown-Lake corridors. These areas correspond to low-elevation floodplains and coastal fringes, confirming the spatial correlation between development pressure and flood hazard. Western rural sectors such as Freshwater and Totland showed lower densities (<100 persons per square kilometre) and thus reduced exposure, although the relative isolation of residents there introduces its own logistical challenges for emergency access.

Part 1: Environment Baseline Analysis

The first part of this consultancy assessment focuses on understanding the spatial relationship between natural and human systems across the Isle of Wight. A comprehensive map (Figure 1) integrates soil classification, population distribution, average rainfall, and primary road networks to provide an overview of the island's physical and social geography.

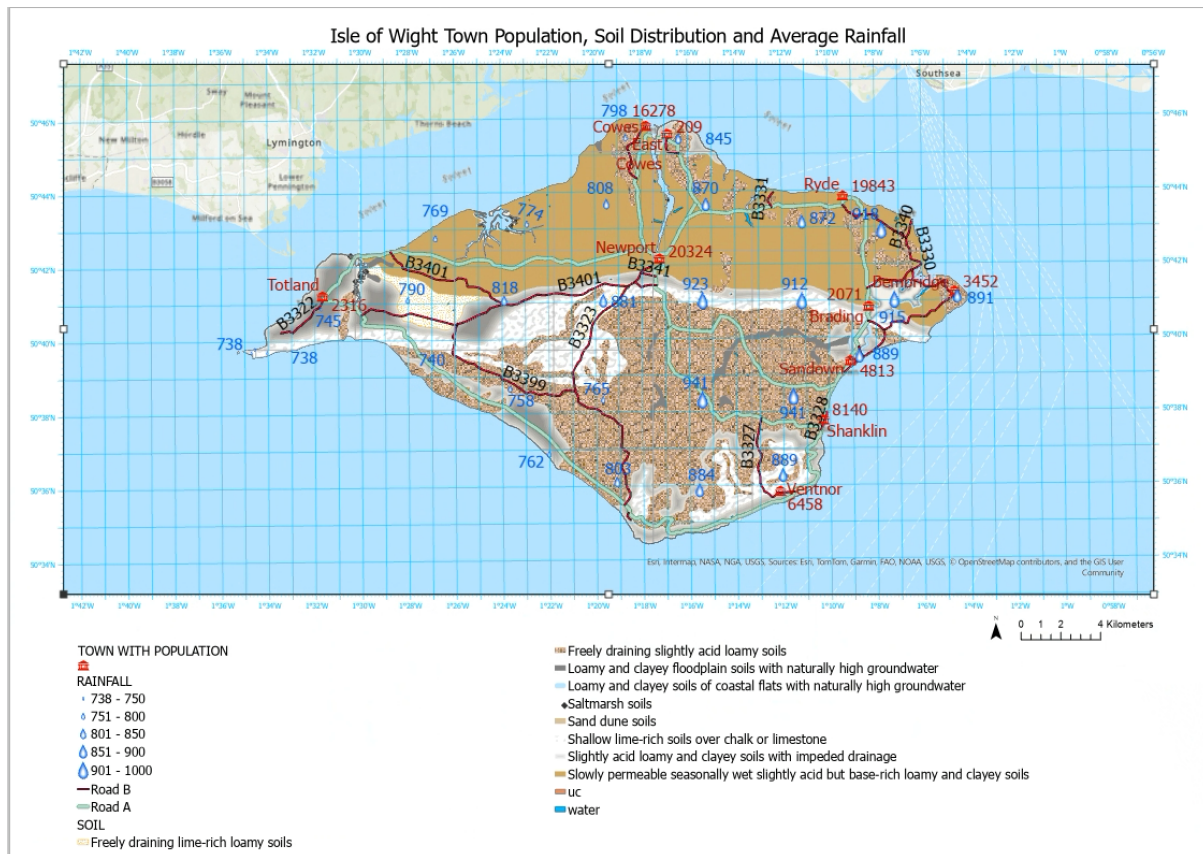


Figure 1- Isle of Wight: Primary Towns and their Population, Soil Distribution and Average Rainfall

1.1 Methodology

Data sources include the Emperian databases along with supporting datasets provided by Cranfield University. Each dataset was reprojected to the British National Grid (EPSG:27700) for spatial consistency. Relationships among soil type, topography, rainfall, and settlement distribution were spatially analysed, and maps were created with legible legends and scales.

1.2 Results and Interpretation

The baseline analysis revealed distinct spatial trends. Freely draining loamy soils dominate the northern and eastern regions, supporting urban centres such as Newport, Ryde, and Sandown. The central valley exhibits clayey floodplain soils, while chalky uplands occur in the south. Rainfall patterns show that a higher precipitation (up to 950 mm/year) occurs over the southern downs and lower values (around 730 mm/year) occur along northern coasts. The alignment of the main road network (A3020, A3056) reflects accessibility between these environmental zones. Population clusters correlate with areas of favourable soils and lower rainfall, confirming environmental influence on spatial development.

Part 2: Demographic Vulnerability in Flood-Susceptible Areas

Part 2 extends the GIS analysis by integrating flood risk data with demographic profiles to assess age-based vulnerability. The derived map (Figure 2) visualises concentrations of children (0-15 years) and older adults (65 years and above) residing within flood-susceptible zones delineated by river and coastal flood boundaries.

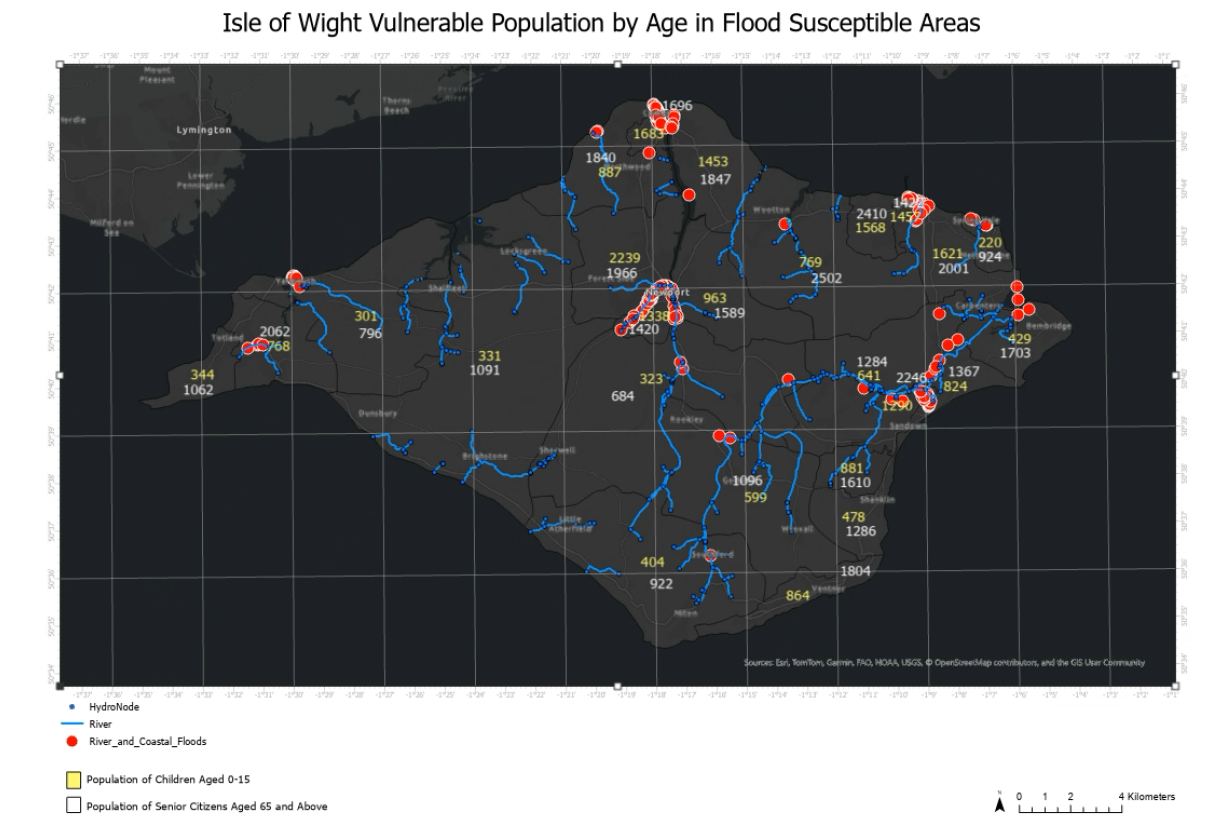


Figure 2 - Isle of Wight: Vulnerable Population by Age in Flood-Susceptible Areas

2.1 Methodology

The Perils Flood dataset, containing riverine and coastal flood polygons, was joined with the R60 Geopoint data using postcode keys or common identifiers. Spatial joins were applied to link vulnerable population attributes to flood exposure zones. Points representing population counts for the two age groups were symbolised using yellow numbers for children and white numbers for senior citizens. The resulting map allows visual interpretation of spatial exposure levels and identifies localities with compounded demographic vulnerability.

2.2 Results and Interpretation

Spatial analysis indicates that Newport, Ryde, and Sandown exhibit the highest dual vulnerability, i.e. large numbers of children and older adults residing within flood-susceptible catchments. Western settlements like Totland and Freshwater show smaller populations but notable exposure along low-lying floodplains. Southern upland communities such as Ventnor and Shanklin show reduced flood risk due to elevation. The overlap of both vulnerable groups in densely populated zones reflects the combined social and environmental challenge of managing flood risk in ageing and dependent populations.

Integrated Discussion

The integration of Parts 1 and 2 demonstrates how environmental conditions and demographic structures interact to define spatial vulnerability on the Isle of Wight. Favourable soil and rainfall conditions have historically driven settlement into the same lowland zones that now exhibit higher flood exposure. This paradox underscores the need for spatial planning policies that balance accessibility and development potential against long-term environmental resilience. GIS analysis reveals that high-risk zones align with ageing populations, implying that socio-environmental risk is compounded by demographic transition.

By combining environmental baselines with hazard and demographic data, GIS provides a comprehensive spatial decision framework. The approach allows planners to quantify vulnerability, prioritise interventions, and visualise complex relationships in an intuitive format. This integrative method is directly applicable to flood management, emergency planning, and sustainable development strategies.

Recommendations

Based on the combined analysis, GeoM Solutions Consultancy proposes the following recommendations to enhance flood resilience and sustainable planning on the Isle of Wight:

1. Maintain GIS-based vulnerability monitoring - Establish a regularly updating GIS database linking flood risk, population demographics, and infrastructure to track spatial change and inform planning.
2. Prioritise high-risk communities – Focus mitigation and adaptation resources on Newport, Ryde, and Sandown, where both hazard and vulnerability levels are greatest.
3. Integrate green infrastructure and Sustainable Drainage Systems- Apply sustainable drainage systems and ecological buffers in floodplain developments, particularly in clayey catchments with poor infiltration.
4. Target social resilience initiatives – Coordinate with local councils to deliver awareness programmes and evacuation planning tailored for elderly and young populations.
5. Embed vulnerability indicators into land-use planning – Use age-based and environmental data layers as constraints during spatial zoning and housing allocation.
6. Continue academic-municipal collaboration – Partnership between Cranfield University and the Isle of Wight Council to refine GIS models incorporating sea-level rise and climate change projections.

Implementing these actions will ensure evidence-based decision-making that reflects the island's demographic realities and environmental sensitivities. The Isle of Wight thus represents a model of how integrated GIS analysis can support adaptive coastal management and community resilience planning.

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Appendix A

Figures and Maps

Figure 1

Isle of Wight – Primary Towns, Soil Distribution, and Average Rainfall

This figure presents a composite environmental baseline map integrating primary towns, soil types, and mean annual rainfall. It provides spatial context for settlement patterns relative to environmental conditions.

Figure 2

Vulnerable Population by Age in Flood-Susceptible Areas

This map illustrates concentrations of vulnerable age groups (children aged 0–15 years and adults aged 65 years and above) within riverine and coastal flood-risk zones.

All figures were produced using ArcGIS Pro 3.5.2 with standardised symbology, labelled scale bars, and north-arrow orientation. Each figure adheres to Cranfield University's GIS map-presentation standards and APA 7 visual-display guidelines, ensuring clarity, accessibility, and consistency in cartographic representation.

Appendix B

Excel-Based Density Calculation Example

This appendix outlines the method used to calculate population density within flood-susceptible areas using Microsoft Excel. The flood-zone area (in km²) was first derived from the Perils dataset using the *Calculate Geometry* tool in ArcGIS Pro. The total population within each zone was then summed by postcode using the Excel formula =SUMIF(range_flood, "Yes", range_pop). Population density was calculated by dividing the total population by the flood-zone area using =B2/A2. For instance, a flood area of 4.8 km² containing 1,152 residents yields a density of 240 persons per km². This calculation provided a straightforward metric for identifying areas with higher demographic exposure to flooding.

Appendix C

Technical limitations of GIS

Several technical limitations were encountered during the implementation of GIS-based analysis for the Isle of Wight vulnerability assessment. Differences in spatial resolution and coordinate systems between the Cranfield-provided Experian datasets and the EDINA Digimap River data created alignment issues that required careful re-projection and attribute standardisation before spatial integration. The non-spatial Perils and Mosaic tables lacked consistent, unique identifiers, which complicated postcode-based joins and occasionally resulted in null attribute values. ArcGIS Pro also restricts the creation of additional attribute indexes for datasets already containing spatial indexes, limiting optimisation of join performance. Furthermore, the large size of the combined geodatabases caused reduced system responsiveness during rendering and geoprocessing operations. Finally, the proprietary nature of the Experian datasets limited verification and comparison with independent open data sources, slightly constraining transparency and reproducibility. Despite these challenges, the overall spatial analysis was completed through data cleaning, projection, and validation of key outputs against Excel-derived summaries.

Appendix D

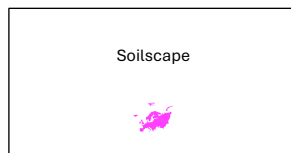
This appendix provides a snapshot of an Excel sheet containing the Metadata for the data used to produce this consultancy report.

1	2	3	4	5
Identifier	Element name	Entry		
1	1 Title	Integrated GIS Analysis of Environmental and Demographic Vulnerability on the Isle of Wight		Mandatory
2	2 Alternative title	Isle of Wight Flood Vulnerability GIS Dataset		
3	3 Dataset language	English		
4	4 Abstract	This dataset represents an integrated spatial analysis of flood susceptibility and population vulnerability for the Isle of Wight. Using Experian Perils flood polygons, R60 Geopoint postcode centroids, and demographic data provided by Cranfield University, population exposure and density were modelled within flood-affected areas. The analysis supports flood-management planning, infrastructure resilience assessment, and community preparedness initiatives.		Mandatory
5	5 Topic category	Environment; Society; Hazard management; Geographic Information Systems		
6	6 Keyword	Flood risk; Population vulnerability; GIS analysis; Isle of Wight; Perils; Experian; R60 Geopoint; Cranfield University		Mandatory
7	7 Temporal extent	Data sources cover 2015 – 2023 (Ordnance Survey 2015; Experian 2023; MOSAIC 2023)		Mandatory
8	8 Dataset reference date	15/03/2025		Mandatory
9	10 Lineage	All data were provided by Cranfield University for academic analysis. Experian Perils and MOSAIC datasets (2023) represent proprietary socio-economic and hazard-risk information. The EDINA Digimap rivers layer (2023) was used for hydrographic context. Data were projected to British National Grid and integrated within a File Geodatabase.		Mandatory
10	15 Extent	Data sources cover 2015 – 2023 (Ordnance Survey 2015; Experian 2023; MOSAIC 2023)		
11	16 Vertical extent information			
12	17 Spatial reference system	British National Grid (OSGB 1936, EPSG: 27700)		Mandatory
13	18 Spatial resolution	1:10,000 – 1:25,000		Mandatory
14	19 Resource locator	https://cranfield.instructure.com/courses/38766/pages/assessment-information		
15	21 Data format	ESRI File Geodatabase (.gdb)		
16	23 Responsible organisation	Cranfield University		Mandatory
17	24 Frequency of update	Not planned		
18	25 Limitations on public access	Restricted – available only to Cranfield University students.		Mandatory
19	26 Use constraints	© Cranfield University 2025; for academic use only.		Mandatory
20	27 Additional information source			
21	30 Metadata date	23/10/2025		Mandatory
22	33 Metadata language	English		
23	35 Metadata point of contact	Manisha Thakur – Cranfield University		Mandatory
24	36 Unique resource identifier	CRANFIELD_GIS_2025_IOW_FLOOD_VULNERABILITY		Mandatory
25	39 Resource type	Dataset		Mandatory
26	41 Conformity			
27	43 Equivalent Scale			
28	44 Bounding box	West – 1.60° W; East – 1.05° W; North – 50.80° N; South – 50.55° N		Mandatory
29				
30				
31	GEMINI v2.2			
32	See http://www.agi.org.uk/uk-gemini/			

Appendix E

Data Structure Diagram: Final Layers in ArcGIS Pro

PART 1



Average Rainfall



Town Names and Population



Roads A and B



Basemap and Graticule



PART 2

Rivers and Hydronodes



River and Coastal Floods



Flood with Population



Basemap and Graticule

